Situation description:

You are a security specialist at a company that processes online payments. The company has recently noticed suspicious activity in its system, which may indicate a possible data breach or cyberattack. Your task is to perform an initial risk analysis and identify potential issues.

Tasks:

1. Identify at least three potential risks that could threaten the company's operations.
2. Assess the impact and likelihood of each risk (low, medium, high).
3. Propose simple preventive measures to reduce these risks.

Risk	Impact	Likelihood	Preventive Measures
DDoS attacks	high	medium	Implement firewalls and DDoS protection mechanisms; ensure high-availability infrastructure.
Phishing attacks	high	high	Provide cybersecurity training for employees and implement multi-factor authentication (MFA). Deploy email filtering systems.
Customer payment data breach	high	medium	Encrypt customer data, implement strict access controls and conduct regular security audits, and ensure PCI DSS compliance to meet mandatory payment industry security standards.
SQL injection / Web application attack	high	high	Use a Web Application Firewall (WAF), enforce input validation and parameterised queries, and conduct regular penetration testing to identify and remediate vulnerabilities.